

Papa Kobina Kwegyir-Aggrey

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Technical Skills

Languages: Python, C/C++, Matlab, SQL (Postgres), Bash, JavaScript/Typescript, HTML/CSS
Frameworks: Tensorflow, Pytorch, OpenCV, ROS2, Scikit-Learn, Numpy, Pandas, Keras, React
Developer Tools: AWS, Git, Docker, Kubernetes, Spark, Linux, Terraform, CloudFormation, Jira (Agile)

Experience

Software engineering Intern

May. 2024 - Aug. 2024

Cooperators

- Engineered and trained an **LLM** chatbot using Retrieval-Augmented Generation (**RAG**) on FAQ data, resulting in **30%** reduction in sales rep engagement time by effectively handling initial user interactions.
- Automated tasks such as Jira issue migration using **REST APIs**, resulting in \$20,000 annual savings by decreasing project lead times and improving overall efficiency.
- Streamlined the policy processing workflow by deploying scalable and reliable **AWS Lambda functions**, resulting in a 15% reduction in processing time.

AI / Machine Learning Intern

Sep. 2023 - Dec. 2023

EON Media

- Orchestrated a scalable **AWS** ML data pipeline using **Apache Airflow** and **Docker** for efficient metadata extraction from **500k+** videos, resulting in a **\$15k/month** cost reduction through improved video search.
- Employed NLP techniques using **OpenAI Whisper model** to transcribe speech-based videos and **Pegasus** to generate concise summaries for team meetings and videos, reducing content review time by **25%** and improving information accessibility for team meetings.
- Redesigned a logo detection model using **YOLOv8**, implementing repeat-learning generated images, achieving an accuracy of **92%**, and reducing dataset preparation time by **40%**.
- Built a face detection model using **OpenCV** and **PyTorch**, achieving **95%** accuracy and reducing inference time by **30%**, enhancing the efficiency of facial recognition processes.

Computer Vision Engineer

Sep. 2023 - Present

Watonomous - Part time

- Enhanced an **autonomous car's** navigation system by implementing a **ROS** node for traffic light detection, utilizing camera data analysis and image processing fine-tuning **YOLOv8** with **Python** to achieve a precision of **93.3%**.

Data Science/ Machine Learning Intern

Jan. 2023 - Apr. 2023

Volta Energy

- Engineered full-stack dashboard to view motor lifespan with **React** and **TypeScript**, reducing downtime by **90%**.
- Developed **deep learning** models using **Tensorflow** to predict motor lifespan and potential faults, resulting in savings of **\$200k**.
- Automated end-to-end pipelines for data collection while enhancing GPU using **CUDA kernel development**.

Projects

AI Voice Assitant | *Python, LangChain, Pydub, Google API Client* | [Link to Project](#)

- Designed an AI-driven voice assistant using Python, enabling voice-based interactions across various tasks using **LangChain Agents** with features such calendar scheduling and auto email response.

NLP Study Buddy | *Transformers (Hugging Face), Streamlit, Selenium, Git* | [Link to Project](#)

- Developed a **multimodal NLP Study Buddy**, integrating text summarization, question answering, and automated flow chart generation.

Lane detection | *OpenCV, NumPy, Python* | [Link to Project](#)

- Implemented Canny edge detection and Hough Line Transform, to perform lane detection with 92% accuracy.

Education

University of Waterloo

Waterloo, On

Candidate for B.A.S.c in Computer Engineering

Sep. 2022 – May 2027

- Relevant courses: Data structures & Algorithms, Digital Computers, Digital Circuits, Linear Algebra